Questions 4(ii), 5 and **6** caused the most problems for even the strongest candidates.

Comments on specific questions

Question 1

For the very first question on this paper candidates' attempts were surprisingly laboured in many cases. Those candidates who knew exactly what to do managed it in less than half a page but very often an answer was not reached despite laboured calculations occupying up to two sides.

One reason for either an incorrect or no solution at all was the fact that many candidates were unable to square a matrix. Some candidates simply squared each number giving no evidence of having tried to multiply two matrices together, whilst others did write the matrix **A** out twice and attempt unsuccessfully, using an incorrect form of row and column multiplication, to multiply out.

Another, more common, reason was the misuse of the identity matrix **I**. Many candidates reversed the zeros and ones whilst others spent a disproportionate amount of time finding and trying to solve the given equation using the inverse of matrix **A** instead of **I**.

Answers: $m = -3$, $n = -5$.

Question 2

Errors made were usually in the manipulation rather than knowledge of cot and cosec. Many candidates appear to need further practice particularly in deciding what they are trying to show and sorting out the terms appropriately. Most solutions started using the left hand side of the identity. Those that started from the right hand side were usually unable to produce the appropriate partial fractions required in the final part of the solution.

When asked to show something as in this question, candidates need to demonstrate the result to the satisfaction of the Examiner. Very often candidates got as far as $\frac{2\cos A}{\sin^2 A}$ and went straight to $2\operatorname{cosec} A \cot A$. In this question, where the candidate is asked to show the given result, an Examiner could not be certain that the candidate understood the appropriate trigonometric relationships without an intermediate step.

Answers: **(i)** $y = 3\sqrt{4x+1}+5$; **(ii)** (0, 15), (30, 0).

Question 7

(i) Most candidates were able to make a reasonable attempt at the use of the given substitution and obtain a quadratic equation in *u*. Many went on to correctly obtain the solution to a power equation, but also tried to solve equations of the type $2^x = k$ with *k* being negative, introducing a spurious solution.

(ii) Many candidates were also able to make a reasonable attempt at this part of the question. Some candidates, although they were able to correctly evaluate $2\log_9 3$ and $\log_2 8$, had problems with the resulting equation and were unable to manipulate the logarithm correctly.

Answers: **(i)** 2.32; **(ii)** 4.

Question 8

Overall, this was a well answered question with many candidates gaining full marks.

(a) Some candidates failed to realize that they had to substitute both $x = 1$ and $x = 2$ into the given expression. However, the vast majority of candidates succeeded in doing this and then multiplied one of the two remainders by 4 to get either 47 (incorrect), or 32 (correct). Most candidates were able to achieve full marks.

(b) The factor $(x + 2)$ was found by most candidates and an attempt made to divide the expression to find the quadratic factor. The most common error at this stage was to find a quadratic factor of $x^2 - 2x + 4$ and then to use the quadratic formula. Most candidates found the correct quadratic factor of $x^2 - 6x + 4$ and then went on to use the quadratic formula to find the other two solutions. However, although many candidates gained the three method marks, the accuracy marks were often lost because firstly candidates omitted to include the root $x = -2$ and secondly they failed to simplify the surd result far enough, even though guidance was given in the question

Answers: **(a)** 32; **(b)** -2, $3 \pm \sqrt{5}$.

Question 9

(i) Most candidates were able evaluate xy and x^2 and draw an appropriate straight line graph. However, there were some candidates who chose to use a non-linear scale on their graph which was deemed to be inappropriate.

(ii) Calculating the gradient of the resulting straight line graph and the intercept on the *y*-axis (or an equivalent method using substitution) proved to be straightforward for most candidates. Problems arose when candidates attempted to rewrite their $y = mx + c$ equation obtained for their straight line graph in the form $xy = mx^2 + c$ and from there to the form requested in the question

(iii) For most candidates finding the value of *y* for which $y = \frac{83}{x}$ proved difficult. Instead of using their graph together with $xy = 83$ and obtaining a value for x^2 and hence *y*, most candidates chose to substitute into an incorrect equation.

Answers: **(ii)** $y = 1.2x + \frac{24}{x}$; **(iii)** a value between 11.6 and 12.2.

Question 10

Nearly all candidates made some attempt at this question and for some weaker candidates it was their best attempted question.

(i) Candidates used various methods, the cosine rule, the sine rule, or basic trigonometry in the form of $2 \times 10 \sin 0.4$, the cosine rule being the most popular method. Premature rounding was a problem, whatever method candidates employed.

(ii) Some candidates caused difficulties for themselves by rounding 1.17 radians to 1.2 radians or by converting to degrees. Others confused degree methods with radian methods and arc length with area. There were some miscalculations of the angle *ABC*. The majority of candidates were able to calculate an arc length, but some candidates were unable to formulate a good plan and included extra parts and some incorrect subtractions. There were, however, many correct answers.

(iii) Many of the same problems that occurred in part **(ii)** also occurred in this part. Many candidates used $\frac{1}{2}r^2(\theta - \sin\theta)$ for either 10 or 7.79 without further work, while some candidates missed the 0.5 from the formula for sector area. There were again, however, many correct solutions.

Answers: **(ii)** 24.9 cm; **(iii)** 39.6 or 39.7 cm^2.

Question 11 EITHER

The great majority of candidates opted to do this alternative.

(i) Most candidates realised that they had to make use of the product rule when differentiating for the first time. Those that were unable to differentiate the exponential term correctly were still able to score for a correct method used, but were ultimately penalised later in the question. The application of the product rule for the second derivative seemed to cause more problems with many candidates omitting terms.

(ii) Most were able to equate the derivative obtained in part **(i)** to 0 and attempt to solve, although many candidates attempted to obtain a solution for the exponential term equated to 0.

(iii) Most candidates were able to apply a valid method for the determination of the type of turning point although there were some who chose to state the type of turning point with no attempt to justify their conclusion.

Answers: **(i)** $e^{2x} + 2xe^{2x}$, $4e^{2x} + 4xe^{2x}$; **(iii)** Minimum.

Question 11 OR

(i) Most candidates realised that differentiation of a quotient was involved and were able to make reasonable attempts at manipulation of the result, although there were some contrived results.

(ii) Most candidates were able to equate the derivative obtained in part **(i)** to 0 and attempt to solve the resulting equation. There were again instances of contrived answers from candidates who were unable to deal with logarithms correctly.

(iii) There were very few correct solutions to this part of the question. Most candidates did not attempt it. Those that did invariably failed to realise the connection with the previous part of the question and produced work of little or no merit.

Answer: **(iii)** $-\frac{1}{4x^2} - \frac{\ln x}{2x^2} + c$ or equivalent.

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